Solution

LIGHT-04

Class 08 - Science

Section A

1. **(d)** translucent

   Explanation: translucent

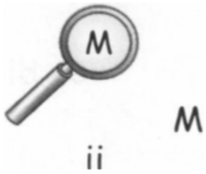
2. **(a)** 45^{o}

   Explanation: 45^{o}

3. **(c)** To control or regulate the amount of coming light to the eye

   Explanation: To control or regulate the amount of coming light to the eye

4. **(b)** ii and iii

   Explanation: 

and 

5. **(b)** 4 cm

   Explanation: 4 cm

6. **(b)** Plane mirror

   Explanation: Plane mirror always forms virtual, erect, and image of the same size. The image is laterally inverted too.

7. **(c)** Plane mirror

   Explanation: Virtual image of same size is formed by plane mirror only. A person want to obtain virtual image of same size should use plan mirror.

8. **(b)** Dispersion of light by water droplets

   Explanation: Rainbow is formed due to the dispersion of light by water droplets after the shower in the opposite direction of Sun. A rainbow is a meteorological phenomenon that is caused by reflection, refraction and dispersion of light in water droplets resulting in a spectrum of light appearing in the sky. Rainbows caused by sunlight always appear in the section of sky directly opposite the sun.

9. **(a)** white

   Explanation: white

10. **(a)** black

    Explanation: black

11. **(a)** Changes with medium

    Explanation: Speed of light changes with medium. Light travels faster in rarer medium and slower in denser medium.

12. **(a)** Kaleidoscope

    Explanation: A kaleidoscope is an optical instrument with two or more reflecting surfaces tilted to each other in an angle, so that one or more objects on one end of the mirrors are seen as a regular symmetrical pattern when viewed from the other end, due to repeated reflection.

13. **(b)** 3

    Explanation: 3

14.

(c) Luminous bodies

Explanation: The object which emits their own light is called luminous bodies. They may be natural or manmade sources of light.

15.

(b) Seven colours

Explanation: The cause of dispersion of light is that white light consists of seven different colours, and each colour has a different angle of deviation. Therefore, on passing through the prism different colours deviate through different angles. Hence the seven colours of white light separate and form a spectrum. Out of seven colours, the red colour deviates the least, and hence the red colour is present at the top of the spectrum. On the other hand, the violet colour deviates most that is why violet colour is present at the lower end of the spectrum.

16.

(d) iris

Explanation: Iris is a dark muscular structure behind the cornea, which controls the amount of light entering into the eye.

17.

(b) Light travels in straight line

Explanation: Light travels in straight line

18.

(d) Real image

Explanation: The image formed on cinema screen is an example of real image as it is obtained on screen. The inverted image is re-inverted by the projector to appear erect.

19.

(d) Reflect or emit rays of light

Explanation: When we see an object when it reflects or emit rays of light. Reflected light enters the eyes and image is formed on the retina.

20.

(d) Converging

Explanation: Converging

21.

(b) Normal

Explanation: Normal is a line drawn perpendicular to the surface of the mirror from the point where incident ray meets the surface of the mirror.

22.

(c) Plane mirror

Explanation: Plane mirror

23. **(a)** Iris

Explanation: The colour of eyes is due to colour pigmentation present in the diaphragm of iris present around the pupil that controls the amount of light entering the eye ball.

24.

(d) Blue coloured light can be scattered by small particles

Explanation: Sky appear blue coloured during day because blue coloured light can be scattered by small particles present in atmosphere due to its shorter wavelength.

25.

(b) Transparent protein

Explanation: Lens transparent body enclosed in an elastic capsule. The lens present in the human eye is made up of proteins and water.

26.

(c) Reflect the rays of light

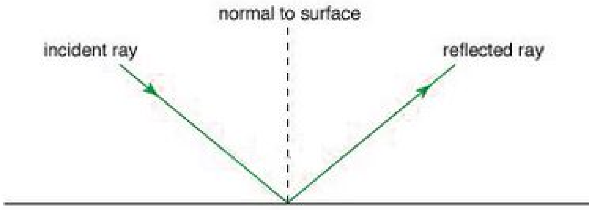
Explanation: Different colours are due to ability of the object to reflect the rays of light. The pigments present in objects absorb some light of the required wavelength and reflect the others.

Section B

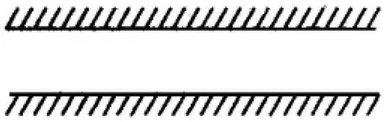
27. A ray of light from a source striking a given surface is called the **incident ray.**

    The point at which the incident ray strikes the surface is called **the point of incidence**.

    Whereas the perpendicular to the surface at the point of incidence is called **normal.**



28. Non-optical aids include visual aids, tactual aids (using the sense of touch), auditory aids (using the sense of hearing) and electronic aids. With the help of visual aids, required size of words, suitable intensity of light and material at proper distance can be provided, in taking notes, reading and writing and in learning mathematics. Auditory aids include cassette, tape recorders, talking books and other such programs. Electronic aids such as talking calculators are also available for performing many computations. Closed circuit television, also an electronic aid, enlarges printed material with suitable contrast and illumination. Nowadays, use of audio CDs and voice boxes with computers are also very helpful for listening and writing the desired text.
29. The size of the eyes of a nocturnal bird is large. The eyes of the nocturnal birds having large cornea with a wider pupil can collect more ambient light which helps them to see the objects even at night.
30. The ray is reflected infinite number of times between the two plane mirrors placed parallel to each other.



31. The impression of an image does not vanish immediately from the retina. It persits there for about $\frac{1}{16}$th of the second. So, if image flash on the eye at a rate faster than 16 per second, then the eye perceives a moving picture of the object. The movies that we see are actually separate pictures. They are made to move across the eye usually at the rate of 24 per second (faster than 16 per second). So we see a moving picture.
32. Braille system has 63 dot patterns or characters. Each character represents a letter, a combination of letters, a common word or a grammatical sign. Dots are arranged in cells of two vertical rows of three dots each. Patterns of dots to represent some English alphabets and some common words are shown below. These patterns when embossed on Braille sheets help visually challenged to recognize words by touching. To make them easier to touch, the dots are raised slightly.
33. **LAWS OF REFLECTION**

    First Law- According to first law of reflection, the incident ray, the reflected ray and the normal at the point of incidence, all lie in the same plane.

    Second Law- According to second law of reflection, the angle of incidence is always equal to the angle of reflection.
34. (a) Polished Wooden Table : Regular reflection will takes place. This is because polished wooden table will have a plane surface.

    (b) Chalk Powder : Diffused reflection. Because the surface of the chalk powder will be uneven.

    (c) Cardboard Surface : Diffused reflection. Since cardboard has a rough surface.

    (d) Marble Floor with Water Spread Over it : Regular reflection as it will act like a plane surface.

    (e) Mirror : Regular reflection . Because mirror has a shiny surface which is even.

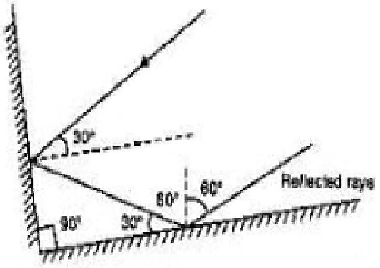
    (f) Piece of Paper : Diffused reflection. Since surface of paper is uneven.
35. Convex lens is present in our eyes, which focuses light on the back of the eye, on a layer called the retina. So, it forms the image of an object at the retina.
36. Lack of vitamin A in foodstuff is responsible for many eye troubles. Most common amongst them is night blindness. One should, therefore, include in the diet components which have vitamin A. Raw carrots, broccoli and green vegetables such as spinach (palak), methi, amarnth and cod liver oil are rich in vitamin A. Eggs, milk, curd, paneer, butter, ghee and fruits such as papaya, banana, mango, apple, dates etc. also contain plenty of vitamin A.
37. Nearly everything we see around is seen due to reflected light. Moon, for example, receives light from the sun and reflects it. That's why we see the moon. These objects are called illuminated.

    There are other objects, which give their own light such as sun, fire, flame of candle electric lamp and others. That's why we see them. These are known as luminous objects.
38. (1) Luminous

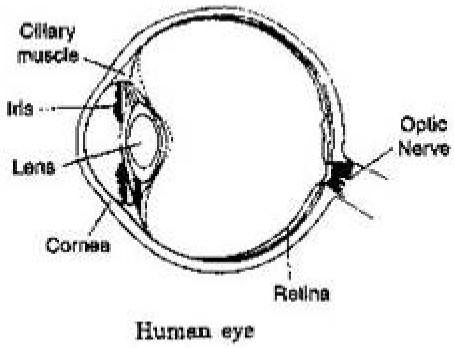
    (2) Single

(3) numerous

39. We can see an object only if light from an object enters our eyes. The light may be emitted by the object or may have been reflected by the object. Thus we cannot see an object which is placed in a dark room if it does not emitt light of its own. Whereas an object outside the dark room can be seen if there is either light outside the dark room on the object emitts its own light.

40. 

41. A kaleidoscope is composed of a hollow tube. Three mirrors are placed in the form of a triangular tube and their reflecting surfaces face each other. One end of the tube is covered with a transparent sheet and another end is covered with an opaque sheet. There is an eyehole in the opaque sheet. Bits of glass are filled inside the tube. When a kaleidoscope is turned, we get to see various patterns in it. These patterns are formed because of multiple reflections.

42. 

Human eye

43. In order to see near objects, muscles attached to the lens contract and the lens becomes thicker. On the other hand, muscles relax and the lens becomes thinner when distant objects are to be seen. This changing of the thickness of the eye lens is called accommodation.

44. In people (particularly old aged) suffering from cataract, the eye lens becomes clouded. Cataract is treated by replacing the opaque lens with a new artificial lens.

45. Characteristics of image formed by a plane mirror are:

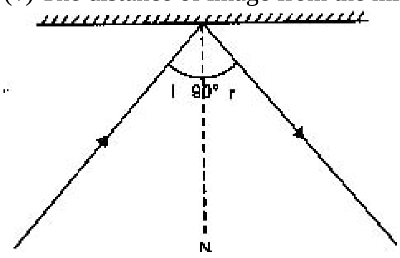
(i) Plane mirror forms virtual images.

(ii) Plane mirror forms erect images.

(iii) Image is laterally inverted.

(iv) Image formed is of the same size as the object.

(v) The distance of image from the mirror is equal to the distance of object from the mirror.

46. 

Given that, $\angle i + \angle r = 90°$

We know that, $\angle i = \angle r$

Replacing $\angle r$ in equation (i) with $\angle i$

$\angle i + \angle i = 90°$

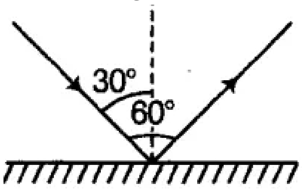
or $2\angle i = 90°$

or $\angle i = \frac{90°}{2}$

$\angle i = 45°$.

47. Light is splitted into its constituent colours when it gets dispersed, e.g. Rainbow formation is due to the dispersion of white light after passing through water droplets.

48. Lateral inversion is the effect produced by a plane mirror in reversing images from left to right. As we observe our image on a plane mirror, we can see that our left side is at the right side of the image and our right side is at the left side of the image. It is also known as perversion.
49. The image of the child cannot be obtained on the screen because the image is not real. The images formed by the plane mirror are virtual, so these virtual images cannot be seen (or obtain) on the screen.
50. Since the angle of incidence = angle of reflection. So, angle of incidence = 30°

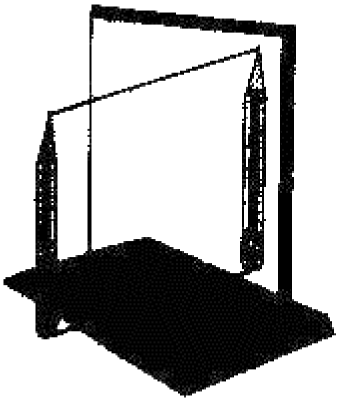


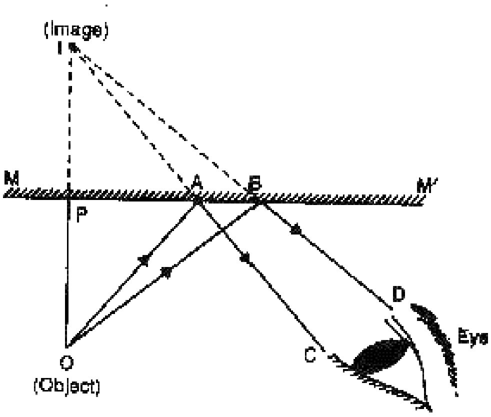
51.

A	B	C	D	E	F
G	H	I	J	K	L
M	N	O	P	Q	R
S	T	U	V	W	X
Y	Z				

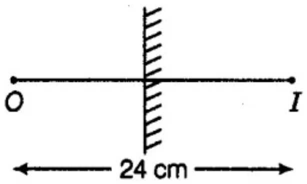
52. Intensity of laser beam is very high, as it carries large amount of energy. It is harmful for eyes and can cause permanent damage. One should not look at laser beam directly or indirectly for a longer period.
53. There are two kinds of nerve endings
(i) cones which are sensitive to bright light and
(ii) rods which are sensitive to dim light. Besides, cones send message of colour. Where the optic nerve leaves the retina, there are no nerve endings. This is called the blind spot.

54.





The image of a pencil formed by a plane mirror is located behind the mirror.
55. Night bird (owl) can see very well in the night but not during the day. On the other hand, day light birds (kite, eagle) can see well during the day but not in the night. The owl has a large cornea and a large pupil to allow more light in its eye. Also, it has on its retina mostly rods and only a few cones. As we have seen above, cones are more sensitive to bright light and rods to dim light. The day birds on the other hand, have more cones and a fever rods.
56. In the case of a plane mirror, the image formed is at the same distance behind the mirror as the object in front of it i.e. object distance from the mirror = image distance from the mirror.



So, the distance between the mirror and the object = 12cm

57. The correct match is:

a : v.

A plane mirror forms an erect image and the size of the image is same as that of the size of the object.

B : ii.

Since a convex mirror forms a virtual, diminished and upright image. Due to which it covers a large area of field of view. That's why a convex mirror can form image of objects spread over a large area.

C : i.

A convex lens is used as a magnifying glass because it forms a magnified image of an object.

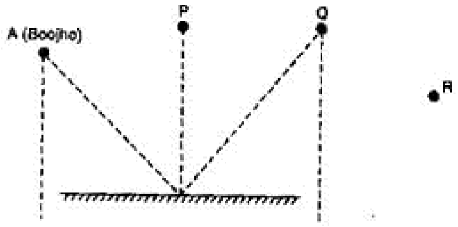
D: iii.

A concave mirror is used by dentists to see enlarged image of teeth because a concave mirror forms an enlarged image of the teeth and is helpful for the dentist.

E:vi.

A concave lens forms a virtual, erect and diminished image i.e. smaller in size as that of object.

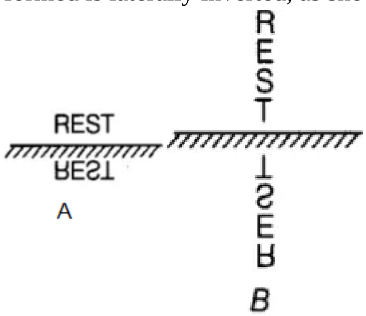
58. No, he can see image of object at P but not of Q and R.



59. Laws of reflection:

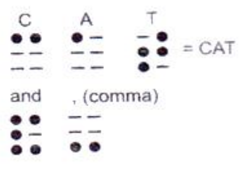
    i. The angle of incidence is always equal to the angle of reflection.
    ii. The incident ray, the reflected ray and the normal to the surface at the point of incidence lie in the same plane.

60. In an image formed by a plane mirror, the left of the object appears on the right and the right appears on the left, i.e. the image formed is laterally inverted, as shown below in figure (A) and (B).



61. Braille is a method of writing for visually challenged person. This system was invented by Louis Braille in 1821 and adopted in 1932. He was also a blind person. In this method text is written on a thick paper using special symbols representing the letters of alphabet. Groups of dots are employed to write letters. There is Braille code for common languages, Mathematics and Scientific notation. Braille system has 63 dots patterns. Each pattern represents a letter, a combination of letters, a common word or a grammatical sign. Dots are arranged in cells of two vertical rows of three dots each. Codes of music and mathematics are different. Any language can be read through the codes of Braille.

Louis Braille



Example of dot patterns used in Braille System

62.

Regular reflection	Diffused reflection

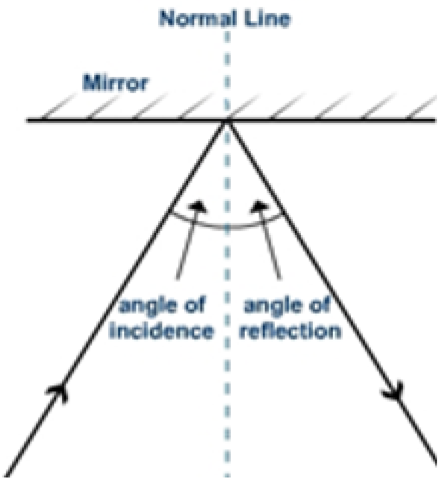
1) Occurs from shiny and smooth surfaces.	1) Occurs from unpolished and rough surfaces.
2) Reflected rays are parallel to each other.	2) Reflected rays are not parallel to each other.
3) Clear images are formed .	3) Blurred images are formed .

The laws of reflection are followed in every situation. Hence, diffused reflection does not mean the failure of the laws of reflection.

63. a. Regular reflection because mirror has smooth surface
    b. Diffused reflection because cardboard has irregular surface.
    c. Diffused reflection, because a piece of paper appears smooth but it has lot of minor irregularities.
64. It is necessary that we take proper care of our eyes. If there is any problem we should go to an eye specialists. Have a regular checkup. We must :
    (i) If advised, use suitable spectacles.
    (ii) Too little or too much light is bad for the eyes. Insufficient light causes eyestrain and headaches. Too much light, like that of the sun, or powerful lamps, can injure retina, which is very delicate.
    (iii) Do not look at the sun or a powerful light directly.
    (iv) Never rub our eyes if any small particles or dust goes into our eyes. Wash our eyes with clean water. If condition does not improve, go to a doctor.
    (v) Wash our eyes frequently with clean water.
    (vi) Always read at the normal distance for vision.
65. First law: The angle of incidence is always equal to angle of reflection.
    Second law: The incident ray, the normal to the surface at the point of incidence and the reflected ray, all lie in the same plane.
66. A clear image is formed by plane mirror due to regular reflection. The image formed by the shiny metal is not clear as that formed by a mirror because metal surface is not so smooth and the refracted rays of light gets diffused.
67. There are three plane mirror strips in kaleidoscope .They are inclined at 60 degree angle to one another forming a hollow prism. It is used by designers on wall papers and fabrics as well as by artist to get new design.
68.

Sr. No.	**Luminous objects**	**Non Luminous objects**
1.	The objects which emit their own light are known as luminous objects. They are also called source of light.	The objects which do not emit their own light are known as Non luminous objects.
2.	Examples: The Sun, fire, flame of a candle and an electric lamp.	Examples: Table, chair, the Moon, the planets, a tree, etc.

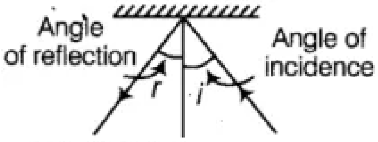
69. a. **Angle of incidence:** The angle between normal and incidence ray is called angle of incidence. Angle of incidence is always equal to angle of reflection.
    b. **Angle of reflection:** The angle between normal and reflected rays is called angle of reflection.

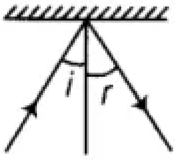


70. Uses of plane mirrors:
    i. We commonly use plane mirrors in our homes to look at our reflection. In beauty parlours, plane mirrors kept at an angle are used to view the side of the head. Plane mirrors parallel to each other are used to view the back of the head.
    ii. Plane mirrors are used to reflect light on an object. For example, during outdoor shooting of a film, metal sheets are used as plane mirrors to reflect sunlight on the actors.
    iii. They are also used in solar cookers to reflect light on the food being cooked.

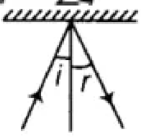
iv. They are used in periscopes. From a submarine under the sea, a sailor can see objects and enemy ships on the surface of the sea by using a periscope.

71. The names of the parts of the eye as shown in the figure are:
    1. Ciliary muscle
    2. Iris
    3. Lens
    4. Cornea
    5. Retina
    6. Optic nerve
72. The correct diagrams are as given below:

a. 

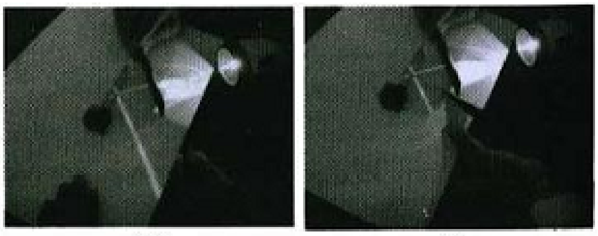
b. 

$\angle i = \angle r$

c. 

Normal should be drawn at 90° to the reflecting surface of a mirror.

73. The non-luminous object can be seen only when light coming from a luminous object falls on them. This light is reflected by the non-luminous objects in all directions. And when this reflected light enters our eyes, we can see the non-luminous object. This is because to us light appears to be coming from the non-luminous objects. Thus we can see the non-luminous object. Because they reflect light received from a luminous object into our eyes.
74. First of all we take a chart paper or stiff paper. Let the sheet project a little beyond the edge of the table.



(a) (b)

Fig (a), (b)

Incident ray, reflected ray and the normal at the point of incidence lie in the same plane

Perform activity again. This time use a sheet of stiff paper or a chart paper. Let the sheet project a little beyond the edge of the Table.

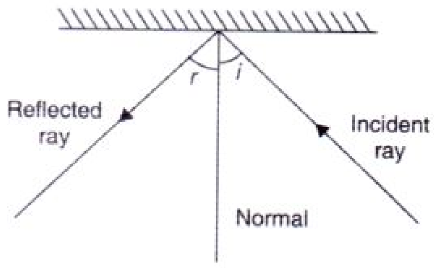
Cut the projecting portion of the sheet in the middle. Look at the reflected ray. Make sure that the reflected ray extends to the projected portion of the paper. Bend that part of the projected portion on which the reflected ray falls. Bring the paper back to the original position. When the whole sheet of paper is spread on the table, it represents one plane. The incident ray, the normal at the point of incidence and the reflected ray are all in the plane when you bend the paper you create a plane different from the plane in which the incident ray and the normal lie. They do not see the reflected ray. What does it indicate? It indicates that the incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane. This is another law of reflection.

75. Reflection is a phenomenon in which a beam of light falls on some surface and returns back in different directions. It may be regular or irregular.

Following are the laws of reflection:

(i) When a ray of light falls on a reflecting surface it is reflected back in such a way the angle of incidence is equal to the angle of reflection, i.e.$\angle i = \angle r$

(ii) The incident ray, the normal and the reflected ray, all lie in the same plane.



Angle of incidence and angle of reflection.